



Al-Ayen Iraqi University
College of Engineering
Department of Artificial Intelligence Engineering
Lab.1 Machine learning
Asst. Lecturer Zahraa Awad



Introduction

Machine learning is the science (and art) of programming computers so that they can learn from data and make predictions or decisions without being explicitly programmed. In this course, we will explore machine learning by leveraging the **Python programming language** along with its powerful libraries, which provide ready-to-use tools for data analysis, visualization, and building predictive models.

Libraries (also called *packages*) are among the most powerful features of the Python programming language. They greatly extend Python's capabilities and make it suitable for a wide range of applications such as data analysis, machine learning, artificial intelligence. Libraries provide pre-written functions and tools that allow developers and researchers to perform complex tasks easily and efficiently without writing code from scratch. ✓

Package Management with pip

pip (short for *Python Installer Package*) is the default package manager for Python. It allows users to install, update, and uninstall external libraries directly from the command line interface (CLI) of the operating system.

Essential Python Libraries for Machine Learning

we will work with four essential libraries, each playing a fundamental role in data analysis and machine learning workflows:

1. NumPy

- A core library for numerical computing in Python.
- Provides powerful tools for working with arrays and performing high-performance mathematical operations. اعتماد أساسي
- Serves as a foundational dependency for many other data science libraries.

2. Pandas تعدد الاستخدامات

- A versatile library for data manipulation and analysis.
- Offers efficient data structures like DataFrame for organizing, cleaning, and analyzing data.

- Makes it easy to read data from various file formats (CSV, Excel, etc.) and perform statistical operations.

3. Matplotlib

- A widely used library for data visualization. ^{نقطي الأعمدة}
- Enables the creation of line charts, bar plots, scatter plots, and many other types of visualizations.
- Helps in interpreting data by presenting it visually.

4. Scikit-learn

- A powerful machine learning library built on top of NumPy, SciPy, and Matplotlib.
- Provides easy-to-use tools for implementing classification, regression, clustering, and other ML algorithms.
- Includes utilities for model evaluation and selection. PCA
confusion matrix & accuracy

Steps to Install the Libraries

1. Opening the Command Line Interface (CLI) : On Windows, the Command Line Interface is known as PowerShell or Command Prompt (CMD).
2. Use the pip install Command : To install the four libraries simultaneously, use the following command:
pip install numpy pandas matplotlib scikit-learn

```

Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\i> pip install numpy pandas matplotlib scikit-learn
Requirement already satisfied: numpy in c:\users\i\appdata\local\programs\python\python313\lib\site-packages (2.2.3)
Requirement already satisfied: pandas in c:\users\i\appdata\local\programs\python\python313\lib\site-packages (2.3.2)
Requirement already satisfied: matplotlib in c:\users\i\appdata\local\programs\python\python313\lib\site-packages (3.10.6)
Requirement already satisfied: scikit-learn in c:\users\i\appdata\local\programs\python\python313\lib\site-packages (1.7.1)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\i\appdata\local\programs\python\python313\lib\site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\users\i\appdata\local\programs\python\python313\lib\site-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in c:\users\i\appdata\local\programs\python\python313\lib\site-packages (from pandas) (2025.2)
Requirement already satisfied: contourpy>=1.0.1 in c:\users\i\appdata\local\programs\python\python313\lib\site-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in c:\users\i\appdata\local\programs\python\python313\lib\site-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\i\appdata\local\programs\python\python313\lib\site-packages (from matplotlib) (4.59.2)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\i\appdata\local\programs\python\python313\lib\site-packages (from matplotlib) (1.4.9)
Requirement already satisfied: packaging>=20.0 in c:\users\i\appdata\local\programs\python\python313\lib\site-packages (from matplotlib) (25.0)
Requirement already satisfied: pillow>=8 in c:\users\i\appdata\local\programs\python\python313\lib\site-packages (from matplotlib) (11.1.0)
  
```

This command downloads and installs the latest versions of the libraries from the Python Package

3. Once the installation is complete, you can verify that the libraries were successfully installed by importing them into a Python script or the IDLE environment:

Libraries.py C:\Users\i\Desktop\Python_work\libraries.py (1.12.5)

File Edit Format Run Options Window Help

```
import numpy
import pandas
import matplotlib
import sklearn
```

```
print("All libraries have been successfully installed!")
```

Python Shell 3.13.5

File Edit Shell Debug Options Window Help

Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win32

Enter "help" below or click "Help" above for more information.

>>>

```
===== RESTART: C:\Users\i\Desktop\Python_work\libraries.py =====
All libraries have been successfully installed!
```

>>>

For example, suppose you want to know if money makes people happy.

1. You download the Better Life Index data from the OECD website.
2. You also download GDP per capita data from the World Bank.
3. Then you join the two tables and sort the data by GDP per capita.
4. Let's plot the data for these countries

ML1.py - C:/Users/I/Desktop/Python_work/ML1.py (3.11.5)

File Edit Format Run Options Window Help

```
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
from sklearn.linear_model import LinearRegression
# Download and prepare the data
data_root = "https://github.com/ageron/data/raw/main/"
lifesat = pd.read_csv(data_root + "lifesat/lifesat.csv")
X = lifesat[["GDP per capita (USD)"]].values
y = lifesat[["Life satisfaction"]].values
# Visualize the data
lifesat.plot(kind='scatter', grid=True,
x="GDP per capita (USD)", y="Life satisfaction")
plt.axis([23_500, 62_500, 4, 9])
plt.show()
```

Figure 1

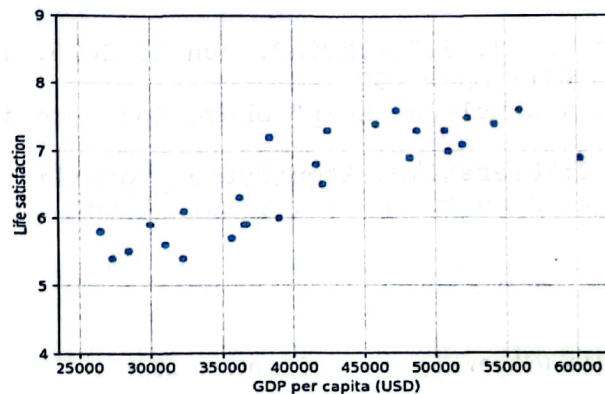


Figure 1.1 :Relationship between GDP and Life Satisfaction

Although the data is noisy, it looks like life satisfaction increases more or less linearly as the country's GDP per capita increases. So you decide to model life satisfaction as a linear function of GDP per capita. This step is called model selection: you selected a linear model of life satisfaction with just one attribute, GDP per capita (Equation 1-1).

Equation 1-1. A simple linear model

$$\text{life_satisfaction} = \theta_0 + \theta_1 \times \text{GDP_per_capita}$$

$wX + b$
تعريف

This model has two model parameters, θ_0 and θ_1 . By tweaking these parameters, you can make your model represent any linear function.

Before using the model, you need to define the parameter values θ_0 and θ_1

How can you know which values will make your model perform best?

you need to specify a performance measure. You can either define a **utility function (or fitness function)** that measures how *good* your model is, or you can define a **cost function** that measures how *bad* it is. For linear regression problems, people typically use a **cost function** that measures the distance between the linear model's predictions and the training examples; the objective is to minimize this distance!

The linear regression algorithm takes the training examples and finds the parameter values that **fit the data best**.

This is called *training* the model. In our case, the algorithm finds that the optimal parameter values are $\theta_0 = 3.75$ and $\theta_1 = 6.78 \times 10^{-5}$.

القبارصة #Cypriots are people from or of the island of Cyprus, which is home to distinct ethnic communities

Suppose you want to know how happy Cypriots are, but the OECD data does not provide the answer.

You can use your trained linear model to make a prediction:

#The OECD (Organisation for Economic Co-operation and Development) is a forum and knowledge hub for data, analysis and best practices in public policy.

1. Look up Cyprus's GDP per capita = \$37,655.
2. apply your model and find that life satisfaction is likely to be somewhere around $3.75 + 37,655 \times 6.78 \times 10^{-5} = 6.30$.

*ML1.py - C:\Users\A\Desktop\Python_work\ML1.py (3.13.5)

File Edit Format Run Options Window Help

```
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
from sklearn.linear_model import LinearRegression
# Download and prepare the data
data_root = "https://github.com/ageron/data/raw/main/"
lifesat = pd.read_csv(data_root + "lifesat/lifesat.csv")

X = lifesat[["GDP per capita (USD)"]].values
y = lifesat[["Life satisfaction"]].values
# Visualize the data
lifesat.plot(kind='scatter', grid=True,
x="GDP per capita (USD)", y="Life satisfaction")
plt.axis([23_500, 62_500, 4, 9])
plt.show()
# Select a linear model
model = LinearRegression()
# Train the model
model.fit(X, y)
print("theta_0 =", model.intercept_)
print("theta_1 =", model.coef_)
# Make a prediction for Cyprus
X_new = [[37_655.2]] # Cyprus' GDP per capita in 2020
print("Life Satisfaction in Cyprus =", model.predict(X_new)) # output: [[6.30165767]]
```

```
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:\Users\i\Desktop\Python_work\ML1.py =====
00 = [3.74904943]
01 = [[6.77889969e-05]]
Life Satisfaction in Cyprus = [[6.30165767]]
>>>
```

Assignment

Example 1-1 shows the Python code that loads the data, separates the inputs X from the labels y, creates a scatterplot for visualization, and then trains a linear model and makes a prediction.